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United States Patent Application Publication

20250282997

Kind Code

A1

Publication Date

September 11, 2025

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Reaction Vessel for Liquid Phase Catalytic Pyrolysis of Polymers

Abstract

An improved reactor vessel for processing (recycling and upcycling) of plastics includes a shear plate assembly disposed on a rotatable shaft concentrically disposed within a cylindrical reactor vessel. The shear plates are spaced from the vessel wall to generate a predetermined strain rate between the shear plate and the inner wall via Couette flow, enhancing efficiency of the reaction.

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Family ID: 1000008626860

Appl. No.: 19/219377

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 17916983 20221004 parent-grant-document US 12312539 US continuation
PCT/US21/26259 20210407 child US 19219377
us-provisional-application US 63006540 20200407

Publication Classification

Int. Cl.: C10B47/36 (20060101); B01J19/18 (20060101); C10B47/34 (20060101); C10B53/07
(20060101); C10B57/06 (20060101)

U.S. Cl.:

CPC C10B47/36 (20130101); B01J19/1887 (20130101); C10B47/34 (20130101); C10B53/07
(20130101); C10B57/06 (20130101);

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/916,983, filed Oct. 4, 2022, which is a 371 national stage filing of International Application No. PCT/US2021/026259, filed Apr. 7, 2021, which claims the benefit of the priority of Provisional Application No. 63/006,540, filed Apr. 7, 2020, each which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to a method and apparatus for pyrolysis of polymers with improved economy and energy efficiency.

BACKGROUND

[0003] In modern life, plastics are ubiquitous. Plastics are inexpensive, easy to mold, and versatile. Plastic properties have many advantages that make them desirable for commercial applications, however, the adverse effects of plastic waste on the environment, especially marine biota, are becoming increasingly apparent. Wastepaper and plastic, mostly packaging materials, are an environmental and public health problem in developing nations where municipal waste pick-up and land-fill disposal are rare or non-existent. Globally, plastic production was estimated to be 380 million tons in 2018. From 1950 to 2018, plastics of about 6.3 billion tons were produced worldwide, 9% and 12% of which were recycled and incinerated, respectively. It has been suggested by researchers that by 2050, oceans might contain more plastics than fish in terms of weight. Yearly, approximately 500 billion plastic bags are used out of which an estimated 13 million tons ends up in the ocean, killing approximately 100,000 marine lives. Recent studies have reported that microplastics and nanoplastics coming from plastic pollution can be found in nearly every organ of the human body and are suspected of playing a possible role in cancers and neurodegenerative conditions.

[0004] Metals, especially aluminum are gathered and recycled because there is a market for them. Unfortunately, recycling of plastics remains a major challenge. There are both technological and economic issues that act to limit the progress in plastics recycling. A slower development within the field of plastics recycling creates a serious problem where millions of metric tons of used polymeric materials are discarded every year around the globe. The fact that the vast majority of used plastics goes to incineration, landfills, or dumps, means that these materials are lost forever as a resource, despite plastics' potential for reuse and recycling. Plastics production requires substantial capital investment and a substantial carbon footprint. Reusing plastics would not only reduce these investment needs but can also contribute to reducing total industrial carbon emissions.

[0005] Many common types of thermoplastics such as polyolefins (HDPE, LDPE, LLDPE, PP) and poly-aromatics (PS, EPS) can be easily separated using commercially available density-based separation methods. While mechanical recycling techniques are effective for some of these plastics, it cannot currently handle mixed polymer streams.

[0006] Chemical recycling uses a chemical reaction called "pyrolysis" which includes cracking of chemical bonds of thermoplastic polymers to hydrocarbon gaseous and liquid products. Pyrolysis is an important process to handle mixed polymers as well as polymers that have exhausted their potential for further mechanical recycling. The energy required for sustaining a pyrolysis reaction is high due to elevated temperatures in range of 430-550° C. and extended reaction residence time. Furthermore, many cracking processes call for the use of hazardous chemicals, which detracts from the intended environmental benefits of recycling.

[0007] In addition to the process' high energy consumption, polymer cracking also produces a lot of carbon that can build up on the bottom of the cracking tower, interfering with the mixing and efficiency of the heat exchange. The rate at which larger molecules are cracked is limited by the

rate at which they can be brought into physical contact with the catalyst surface. Since the mean free path between molecular collisions is two to three orders of magnitude greater in the vapor than in the liquid phase and higher temperature implies greater thermal kinetic motion, the turbulence found in the fluidized bed of a crude petroleum cracking tower (where very hot catalyst particles are blown together with hot petroleum vapor) is sufficient to sustain a satisfactory reaction rate without resorting to mechanical mixing (rotating paddles, etc.) which is necessary in the liquid phase.

SUMMARY

[0008] The present invention is directed to a solution for several of the principal disadvantages of the pyrolysis (cracking) of large organic molecules (aka catalytic de-polymerization or CDP), particularly polymers in a hydrocarbon solvent with the aid of heterogeneous (solid) catalysts. Specifically, the inventive process provides an improved method of supplying heat to the reaction vessels. It also improves the manner by which heat is transmitted into the reactants, increases the efficiency of mixing of the reactants and catalyst and facilitates handling of elemental carbon by-product and un-reactive solids in the feed stock.

[0009] According to embodiments of the invention, a “shear plate reactor” is provided comprising a cylindrical barrel with rotating arms that guide “shear plates” or blades close to the reactor wall such that very high shear strain rates are produced. This mixing occurs at the surface where heat needed for endothermic reactions enters the reactor.

[0010] In some embodiments, the general configuration and dimensions of shear plate reactors, and the significant enhancement the new shear plate reactor design provides to the efficiency and effectiveness of a multi-reactor facility, makes possible the creation of CDP plants with capacities of 100 tons/day and higher.

[0011] In one aspect of the invention, a reactor vessel for liquid phase pyrolysis, includes a cylindrical reactor barrel having a barrel diameter and a barrel length; a heat exchange shell surrounding the reactor barrel, the heat exchange shell connected to a heat exchange circuit and configured to flow a heat exchange fluid over an outer surface of the reactor barrel; a reactor top plate disposed at a top of the reactor barrel; a reactor bottom plate disposed at a bottom of the reactor barrel; a central shaft supported concentrically and rotatably within the reactor barrel; at least one arm set extending radially from the central shaft, each arm having a distal end configured to support a shear plate extending parallel to the central shaft at a spacing from an inner wall of the reactor barrel, the spacing configured to generate a predetermined strain rate between the shear plate and the inner wall via Couette flow; a drive motor supported by the reactor top plate and configured for driving rotation of the central shaft; an inlet for introducing fluid into the reactor barrel for treatment; and at least one outlet for removing processing products from the reactor barrel. Each arm may further support a scraper configured to remove solids deposited on the inner wall, the scraper having a length substantially equal to a length of the shear plate. The at least one arm set comprises a plurality of arm sets spaced apart along a length of the central shaft, and wherein one arm of each of the plurality of arm sets supports a single shear plate having a shear plate length substantially equal to the barrel length. A trailing edge of the shear plate may be curved away from the inner wall to diffuse fluid flow and direct it away from the inner wall. Each arm may include a piston configured to apply variable pressure against the distal end of the arm.

[0012] In some embodiments, an electrically-conductive wire may extend through each arm and be electronically connected at distal end to the shear plate and at a proximal end to a conductivity detector, wherein conductance is measured to indicate physical contact between the shear plate and the inner wall.

[0013] The heat exchange circuit comprises a molten salt circuit which includes solar concentrators. The predetermined strain rate achieved by the shear plates may be on the order of 10×10^2 to 10×10^3 per second. The aspect ratio between the barrel length and the barrel diameter is preferably within a range of 5:1 and 10:1. The drive motor is a hydraulic motor configured to

drive rotation of the central shaft at 40 to 60 rpm.

[0014] In another aspect of the invention, a processing facility includes a plurality of the aforementioned reactor vessels where the reactor vessels are connected in series. Each of the plurality of reactor vessels may be configured to operate at a different processing temperature, where the processing temperatures progressively increase from a first reactor in the series to a final reactor in the series. A plurality of input tanks may be disposed in parallel, each having an outlet in fluid communications with a first reactor vessel, so that material to be processed is pre-processed in alternating input tanks to provide a substantially continuous flow of material into the first reactor vessel.

[0015] In still another aspect of the invention, a method for recycling or upcycling of plastics includes mixing a dispersion of particles of plastic, solvent oil and a catalyst to form a slurry; heating the slurry to a processing temperature; and feeding the heated slurry into the aforementioned processing facility to process the slurry.

[0016] In yet another aspect of the invention, an improved reactor vessel for liquid phase pyrolysis includes a shear plate assembly disposed on a rotatable shaft concentrically disposed within a cylindrical reactor vessel, the shear plate assembly comprising a shear plate disposed on an arm extending radially from the shaft and parallel to an inner vessel wall surface, the shear plate and arms configured to rotate within the vessel to produce a viscous drag between the inner vessel wall surface and an outer surface of the shear plate. The improved reactor vessel may further comprise a scraper is disposed on a leading edge of the shear plate to remove material deposited on the inner vessel wall surface.

[0017] A heat exchange shell may be provided to surround the reactor vessel, where the heat exchange shell is connected to a heat exchange circuit and configured to flow a heat exchange fluid over an outer surface of the reactor vessel. The heat exchange circuit may be a molten salt circuit that includes solar concentrators.

[0018] The arms may be a plurality of arm sets spaced apart along a length of the rotatable shaft, so that one arm of each of the plurality of arm sets supports a single shear plate having a shear plate length substantially equal to a length the reactor vessel. The trailing edge of the shear plate may be curved away from the inner vessel wall surface to diffuse fluid flow and direct it away from the inner vessel wall surface. Each arm may include a piston configured to apply variable pressure against a distal end of the arm. In some embodiments, an electrically-conductive wire extends through each arm and electronically connects at distal end to the shear plate and at a proximal end to a conductivity detector, allowing conductance to be measured to indicate physical contact between the shear plate and the inner vessel wall surface.

[0019] The viscous drag generated by the shear plates corresponds to a strain rate on the order of $10^{2.2}$ to $10^{3.3}$ per second. An aspect ratio between a reactor vessel length and a reactor vessel diameter may be within a range of 5:1 and 10:1. A hydraulic motor attached to the shaft is configured to drive rotation of the rotatable shaft at 40 to 60 rpm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a cross-sectional side view of the upper portion of a reactor head assembly according to an embodiment of the invention.

[0021] FIG. 2 is a diagrammatic top view of a shear plate arm within the interior of a reactor head assembly according to an embodiment of the invention.

[0022] FIGS. 3A and 3B are perspective and exploded views, respectively, of a shear plate arm assembly according to an embodiment of the invention.

[0023] FIGS. 4A and 4B are partial cross-sectional top views of embodiments of a shear plate arm

with a shear blade.

[0024] FIGS. 5A and 5B are a diagrammatic perspective views of the interior of a cylindrical reaction vessel showing the shear plates and scrapers according to an embodiment of the invention.

[0025] FIG. 6 is a cross-sectional view of the bottom of a reactor vessel according to an embodiment of the invention.

[0026] FIG. 7 is a perspective view of the bottom of a reactor vessel according to an embodiment of the invention.

[0027] FIG. 8 is a block diagram of an exemplary preparation sequence for CDP processing.

[0028] FIG. 9 is an exemplary plant schematic incorporating a plurality of reactors according to an embodiment of the invention.

[0029] FIG. 10 is a schematic of a molten salt circuit for use in a conversion plant according to an embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

[0030] While CDP of plastics and crude petroleum are both pyrolysed (i.e., “cracked”) using solid catalysts, petroleum is generally processed as a vapor and at higher temperatures and pressures than are optimum for CDP where a liquid phase is employed to regulate the temperature of the reaction. The improvements disclosed herein are directed to multiple aspects of CDP reaction vessels and processing, each improvement in itself representing a significant advancement toward a more efficient and environmentally-sustainable process for plastics recycling.

[0031] For heat sources, electrical resistance heat is attractive because it is easily controlled and has been applied directly to a reaction vessels in pilot-scale (approximately 1 to 10 Tons/day) plants and indirectly via a heat exchange fluid in an experimental commercial scale (~100 Tons/day) plant where recovery of gas turbine exhaust heat had been proposed but not attempted in practice.

[0032] It has been proposed that mechanical rather than thermal energy can be used to drive the endothermic pyrolysis. Such proposals include the use of impinging jets, high performance (albeit externally heated) mixers, and collapsing cavitation bubbles. U.S. Pat. No. 7,473,348 of C. Koch describes application of electric heat to the walls of the reaction vessels with the additional energy being supplied mechanically through impingement of a high velocity jet upon the counter flowing circulation of a hydro cyclone. A heat exchanger external to a stirred reaction vessel is described in U.S. Pat. No. 9,371,492 of Spitzauer, et al.

[0033] Although electrical resistance heating has been used to supply most of the energy and mechanical (viscous dissipative) heating claimed to provide some additional energy input for pilot-scale testing of CDP, it is generally not economically viable to use these methods on a commercial scale operation because three to five times as much thermal energy must be expended to supply the electric energy required for resistance heat or an electric motor than to apply the thermal energy directly.

[0034] While conversion of electrical energy and shaft work to heat is 100% efficient, conversion of heat to work is limited to some fraction (usually less than half) of Carnot's $\Delta T/T$. The product of this heat to work efficiency together with, electric generation, transmission and motor efficiencies results in an overall efficiency on the order of 20-30%.

[0035] The thermal energy required to raise the temperature of a polymer, e.g., polyethylene, from ambient (20° C.) to the boiling point of diesel (390° C.)

$$[00001] C_p \quad T \sim (2.2 \quad \frac{\text{kJ}}{\text{kg} \cdot \text{K}}) (390 - 20) \text{ K} = 814 \quad \frac{\text{kJ}}{\text{kg}},$$

whereas the energy that must be supplied to cause the endothermic pyrolysis of polyethylene to diesel given the enthalpy of formation of each (from PropEP checked with NIST) is:

$$[00002] H_{\text{pyr}} = H_{f, \text{PE}} - H_{f, \text{D}} = -1.99 - (-2.05) = 58.6 \quad \frac{\text{kJ}}{\text{kg}}$$

so that more than fourteen times as much energy goes to raising the reactant's temperature than to drive the pyrolysis. (Note that the US Navy's Propellant Evaluation Program (PropEP) database values were used for H.sub.f,P E and H.sub.f,D which give larger H.sub.pyr than NIST. (NIST has

H.sub.f for Cetaine C.sub.16H.sub.34 but not diesel).)

[0036] In a CDP reactor, many products of increasing molecular mass and vapor pressure will emerge from the process mixture as it is heated to its final temperature just above the diesel boiling range. The argument about C.sub.p $\Delta T \gg$ H.sub.pyr holds for all products, and when the heats of solution and vaporization are added to the integrated heat capacities, it is evident that electric or mechanical work are not viable energy input mechanisms for large scale pyrolysis reactors.

[0037] A CDP reactor needs to operate just beyond the diesel boiling range of 390° C. and a useful heat transfer fluid must be stable at least 20-30° C. higher than this in order to avoid the economic burden of too large heat exchange surface areas and too high fluid pumping rates.

[0038] Mixtures of Biphenyl and Diphenyl Oxide (BDOM) are perhaps the most thermally resistant organic heat transfer fluid available. One such mixture, Eastman VP-1®, has been used as a heat transfer fluid between electric heaters and heat exchange with the CDP process fluid. (See, e.g., U.S. Pat. No. 9,371,492.)

[0039] The maximum operating temperature of Eastman Thermotrol® 75 is 385° C. while Eastman VP-1 has a slightly higher maximum operating temperature (400° C.) its vapor pressure must be regulated at 10 Bar to remain a liquid at 390° C. While the heat of vaporization of a boiling BDOM is an excellent way to control the temperature of exothermic reactions, it is not suitable as a heat exchange fluid between endothermic pyrolysis that needs to operate near VP-1's T.sub.max and a heat source such as concentrated solar at least 150° C. greater than T.sub.max, turbine or reciprocating engine exhausts $\sim 500^\circ \text{C.} > T_{\text{sub.max}}$, or even hydrocarbon/air flame at $\sim 1000^\circ \text{C.} > T_{\text{sub.max}}$.

[0040] Excursions above set-point temperature in heat transfer circuits can occur due to leaks, failure of valves, pumps, operator error, controller overshoot response to transients, etc. With BDOM boiling liquid and vapor, increased temperature results in increased pressure unless the vapor is cooled by expansion or by heat exchanger/condensers, or preferably both.

[0041] Since leakage or venting of a BDOM to ambient results in a flammable or explosive fog, using them requires a large overpressure safety factor above the 10 Bar operating pressure (vs. ambient pressure plus a modest SF of the CDP process), as well as large expansion tanks and condensers. The commercial-scale test plant described in U.S. Pat. No. 9,371,492 used large VP-1 expansion tanks with a back-up relief vent to ambient.

[0042] Reactants must be brought up to a temperature at which pyrolysis can take place after which a continuous supply of energy must occur to maintain the reaction temperature to promote the mildly endothermic decomposition of polymers into smaller molecules that are useful particularly for transport fuel. Hydrocarbons in general, especially hydrocarbon and polysaccharide polymers, as well as heat exchange oil solvents that can resist decomposition at the high reactor temperatures (up to around 400° C.) have low thermal conductivity, low diffusivity ($\alpha = k/\rho C_{\text{sub.p}}$), and high viscosity, which impede convective and advective heat exchange.

[0043] In U.S. Pat. No. 9,371,492, a heat exchanger external to a stirred reaction vessel were employed with electrically heated VP-1 circulated in the shell side. The process fluid circulated in the heat exchanged tubes, where fouling and plugging were anticipated and experienced in a commercial-scale test plant.

[0044] The reactant polymers need to be brought into proximity of the solid catalyst surfaces by vigorous mixing. The relatively high viscosity of thermally stable hydrocarbon-based solvents as well as the reactant polymers increases the mechanical work to accomplish the necessary mixing.

[0045] Various prior art teachings include that energy to drive the endothermic pyrolysis should come from impinging jets, collapsing cavitation bubbles, and special mixers, such as described in US Patent Publication No. 2008/0116116 of Koch.

[0046] Carbon build-up on heated reaction vessel surfaces and in heat exchanger flow passages has historically been a process problem. Catalyst that has become inactive due to a coating of carbon, i.e., “spent” catalyst, and solid byproducts, mostly elemental carbon at about 10% of the feed mass,

must be removed for further treatment or disposal.

[0047] In patent publications WO2008/061484 and US2008/0116116 of Koch, each incorporated herein by reference, solids (carbon plus unreactable material) are separated by gravity. A subsequent patent, U.S. Pat. No. 7,473,348, also of Koch, removes carbon build-up on process vessel walls with scrapers and from the process fluid with a hydro cyclone. Spent catalyst, along with other un-reacted solids, are heated to 550° C. to burn off the coke so that the catalyst can be returned to the process, as is generally done with petroleum gas-phase cracking.

[0048] According to the inventive approach, shear plates used together with conventional scraper blades are highly effective in addressing the problems arising from each of: 1) slow heat exchange into an oily (viscous, low thermal conductive) liquid; 2) incomplete and slow mixing of reactants with catalysts; and 3) fouling of heat exchanger and reactor surfaces with byproduct carbon. Furthermore, while the physics of reaction energy supplied via mixing work is sound, the economics on a commercial scale is not.

[0049] Cylindrical reactors with rotating scrapers share this feature with scraped surface heat exchangers (SSHE) but differ from SSHE in their use of shear plates to efficiently produce very high strain rates comparable to small, high-speed blade or paddle impellers such as are found in continuous stirred tank reactors. The shear plates do not actually scrape when performing their intended function.

[0050] The Couette flows created by the shear plate reactor disclosed herein differ from those found in Taylor-Couette reactors that have relatively widely-spaced concentric cylinders that generate multiple vortex cells known as “Taylor cells”. The inventive shear plate reactor design employs discrete blades that run very close to the reactor wall to produce the desired high strain rate.

[0051] Temperatures reached by concentrated solar power (CSP) with tracking trough collectors fulfill the requirements of the CDP process. The portion of a CSP array's thermal output invested in CDP results in a form of long-term storable energy that compliments the CSP's shorter duration thermal energy storage. FIG. 10 provides a schematic of an exemplary CSP-heated molten salt circuit 502 for a plastic conversion plant. (See, e.g., salt circuit 502 shown in FIG. 9.)

[0052] It would be particularly desirable to be able to run a heat engine, such as an Ericsson, Brayton, Diesel, Stirling, Otto, Rankin (in order of preference) on the less valuable overhead gasses and less easily-stored distillate fractions, and to utilize the engine's waste heat for the catalytic pyrolysis process. Electric power produced by an engine is best used for the “hotel load”—system components not related to the main power demand of heating, such as pumps, mixers, actuators, controls, etc., with any excess power being sold off-site rather than being used for heating pyrolysis reaction vessels.

[0053] Nitrate molten salt, sometimes referred to as “solar salt” (60% NaNO₃, 40% KNO₃) used in CSP plants, is a good candidate for use as a heat transfer fluid for the CDP process, regardless of whether solar energy is the primary heat source. Molten salts can operate up to 550° C. and do so at ambient pressure. A leak simply solidifies rather than forming an explosive fog like organic heat transfer fluids do. On start-up, provision must be made to bring the process fluid up above the salt melting point (~250° C. for solar salt, but heat treat salts have a melting point as low as 135° C.) in an auxiliary heat exchanger with electric, gas fired, or engine exhaust heat.

[0054] Reaction chamber configuration: Cylindrical (304L SS) reaction vessels have high axial ratios (length/diam from 5:1 to approximately 10:1), so that a reasonable surface to volume ratio is maintained for heat exchange through the walls of the vessels. The maximum diameter is about one meter, but somewhat smaller diameters fall within the range of standard welded tubes for better economy of construction and have better heat exchange surface/volume ratios.

[0055] A rough mass build-up for a reactor with an inner diameter (ID) of 0.761 m and a length of 5 m with an estimated specific gravity of 1.30 [g/cm³] is provided in Table 1 below.

TABLE-US-00001 TABLE 1 OD Specific Mass [in] Wall [in] Len [in] gravity [lbm] 30.0 14.00

177.17 1.30 5852.76 Process fluid charge 34.0 2.00 196.85 2.20 3144.34 Solar salt hx fluid 30.0
0.50 196.85 7.85 2585.76 Reactor barrel 32.0 15.00 2.00 7.85 454.18 Reactor ends 34.0 0.12
196.85 7.85 742.31 Heat exchange shell 38.0 2.00 196.85 0.20 321.58 Insulation 38.0 0.03 196.85
7.85 166.43 Insulation cover 3.0 0.38 98.43 7.85 86.28 Upper scraper shaft 3.0 0.19 98.43 7.85
46.22 Lower scraper shaft

[0056] In Table 1, if typical expected solvent and reactant densities and mass fractions are chosen, the reactor mass build-up would be:

TABLE-US-00002 TABLE 2 Component mass [Ton] Reactant 1.19 Solvent 1.46 HX Salt 1.43
Structure 2.30 Total mass 6.08

[0057] This is an appropriate size and mass for economical fabrication, transport, in-plant installation and replacement or maintenance in an industrial scale facility.

[0058] A cross-section of an exemplary reactor head assembly **100** according to an embodiment of the invention is shown in FIG. **1**. Components are designed for economy of fabrication, maintenance and operation at high temperature. Reactor plate **102** sits on top of cylindrical reaction vessel **104** providing support for drive motor **112**, which rotates shaft **146** within the vessel. Reactor plate **102** may be attached and sealed to the upper plate of vessel **104** by a clamp **138**, such as a Marman clamp to take advantage of their strength and ease of attachment and removal. Alternatively, a bolt ring may be used. Motor **112**, which will typically be a low speed, high torque hydraulic motor (e.g., Eaton H-series Shar-Lynn), is mounted on reactor plate top **103** and connected to upper shaft clamp **124** via a Woodruff key. Upper shaft **114** is enclosed within welded steel frame **115**, which is bolted to top plate **103**. Shaft **114** is guided by one or more bearing **116**, which may be a tapered roller bearing or thrust bearing such as a Glacier-Garlock CBM self-lubricated sintered graphite-bronze. Upper shaft **114** is connected via flange **128** to central shaft **146**. Flange **128** and shaft **114** are insulated by a disk of high density ceramic fiber board or mica from central shaft **146**. The reactor top plate and drive components attached to it are also insulated by a 50 mm (2") layer of closed-cell ceramic foam insulation **123**, which is supported and shielded by 20 gauge stainless steel sheets **127**. All heated components have at least 50 mm-75 mm (2"-3") of heat-stable insulation (T-glass or equivalent) on their outer surfaces, generally indicated by dashed line **139**. Four lugs **121**, arranged radially around reactor plate **102** at 90° spacings allow for removal and replacement of the drive head and shear plate assemblies, lifting of the entire reactor, and/or attachment of sway bracing when reactors are in operation.

[0059] Fluid to be processed is introduced into vessel **104** through inlet port **105**. Vapor exits the vessel at exit port **106**. Inlet **105** and exit **106** are located in the barrel of reactor vessel **104** to facilitate removal of the head and shear plate assemblies. Additional components of the reactor head assembly are shown in the figure but are not separately labelled. These components include seals, glands, bearings, clamps, fasteners, insulation, and other components that are commonly used in such systems. Identification and selection of such components will be within the level of skill in the art and are not described herein.

[0060] Heat exchange shell **136** may be formed from steel, preferably stainless steel, tubes having a radius that is about 5 cm (2") larger than the outer diameter of the reaction vessel to provide passages for heat transfer fluid. To minimize differential thermal expansion between the shell and the reactor wall, shell **136** is preferably made of the same alloy as the reactor vessel **104** (304L). Heat exchange shell **136** is disposed concentrically around cylindrical reaction vessel **140** and bolted via welded flanges to annular flanges on the reactor barrel to provide passages for heat transfer fluid. Steel baffles **137** are brazed to the outer surface of the cylindrical reaction vessel **104** with 45° gaps on alternate sides to guide heat transfer fluid, e.g., solar salt fluid, to all areas of the outer reactor cylinder wall. All heated components have at least 50 mm-75 mm (2"-3") of heat-stable insulation on their outer surfaces.

[0061] Extending radially from central shaft **146** are multiple sets of arms **148**. Central shaft **146** is driven by motor **112**, causing it to rotate the one or more sets of blades at about 40 to 60 rpm that

perform multiple functions. In the illustrated examples, the blade assembly includes a pair of blades supported on arms **148** that are arranged 180° apart on shaft **146**. As will be apparent to those in the art, additional arms and blades may be employed, for example, three blades separated by 120°, or four blades separated by 90°. Accordingly, the two blade arrangement is provided as an illustrative example.

[0062] Referring to FIG. 2, the key elements of each blade are a shear plate **142** and a scraper **143**, which removes carbon (coke, ash) that deposits mostly on the heat transfer surface **141**. Stainless steel shear plate **142**, which is about 5 mm (3/16") thick, is supported so that its surface is substantially parallel to and spaced on the order of 2 to 3 mm away from the reactor vessel wall **141** so that a strain rate on the order of 10.sup.2 to 10.sup.3 per second is maintained by the Couette flow between shear plate **142** and wall **141**. As shown, process fluid flow is diffused by the aft edge of shear plate **142** and scraper **143**, deflected away from the wall **141** and eddy aft of the shear plate assembly.

[0063] The leading edge of scraper blade **143** deflects the process fluid, both newly reacted and as-yet unreacted material) away from reactor wall **141** and toward the central part of the cylindrical reaction vessel while the eddy downstream of the scraper blade advects process fluid from the bulk back toward the heated reactor wall **141**, thus increasing the overall heat exchange rate. Trailing edge **122** of shear plate **142** is curved away from the reactor wall **141** to stiffen the shear plate assembly, to diffuse the flow and, together with the leading portion of the scraper blade **143**, direct it away from the reactor wall, taking advantage of a small thrust in the direction of blade motion resulting from the volume increase upon pyrolysis and evolution of vapor.

[0064] Shear plate arms **148** are welded to central shaft **146** and the joint may be reinforced with an annular gusset **158**. Within each arm, piston **164** moves radially within bronze bushing **165**. The distal end of piston **164** is welded to frame **147**, which carries the axle for wheels **149** (2 per arm). Bracket **151** attaches to frame **147** via three bolts to allow adjustment. Wheels **149** may be bronze-bushed or simply bronze. Scraper **143** is mounted on scraper arm **155**, which is attached to frame **147** via axle **157**.

[0065] Piston **164** is biased outward by spring **162** in chamber **161**. The combination of wheels **149** and piston **164** allow the shear plate **142** to follow small out of round deviations of the reactor barrel or acentric location of the shaft **146**.

[0066] FIGS. 3A and 3B illustrate an embodiment of the shear plate arm with a more robust "truck" assembly than the design shown in FIG. 2. This configuration provides increased support for the shear plate **242** and scraper **243** through the addition of a leading wheel **204**, so that two wheels **204** and **210** straddle shear plate **242**. In order to tolerate greater asymmetry between the central shaft and reactor barrel, a bronze wrist bearing **296** is fitted inside wrist bearing housing **297** that is welded to frame **224**. The wrist pin includes two halves: the first half **295** has a center bore for receiving bolt **294**, while the other wrist pin half **298** is threaded for the bolt that clamps the two wrist pin halves to piston **293**. Piston **293**, which is spring biased, slides within bronze bushing **292** within arm **291**.

[0067] Trailing wheel **210** and leading wheel **204** each run on bronze bushings **203** and are retained axially by large thrust washers **202**. Wheel bushing **203** and thrust washers **202** are clamped to frame **224** by bolt **201** and nut **206**. The weight of the shear plate **242**, scraper **243**, and the truck itself is borne by the bronze thrust bearing **299** that is retained in the upper end of wrist bushing housing **297** with a circlip **210**.

[0068] In an exemplary 5 meter long reactor, a 5 meter long shear plate **242** will typically be supported along its length by five arms. (See, e.g., FIGS. 5A and 5B.) In order for each of the five trucks to bear equal parts of the shear plate's weight, reinforcing bars **214** are provided, tapped for machine screws and tightened to hold the bars to shoe plates **207**, which, in turn are attached to frame **224**. The shear plates, or segments that make up the shear plates, are then welded to the reinforcing bars **214** as the last step of the assembly.

[0069] Scraper **243** is retained on a scraper arm and spring assembly **215**, which extends from a spring **212** formed from a heat resistant alloy, to provide an opposing force on the scraper. The scraper arm pivot is provided with a bronze bushing. This scraper assembly is secured with the same nut and bolt that clamps the trailing wheel **210** to frame **224**. A spring force adjustment bolt **213** reacts via a small bracket welded to the frame.

[0070] FIGS. **4A** and **4B** illustrate additional embodiments of the shear plate assembly that employ flexible shear blades **342**, **343** attached to the distal end of piston **364** by bracket **341**. Shear blades **342**, **343** are made from alloys such as INCONEL®, an austenitic nickel-chromium-based superalloy. The undeflected/free shape of the blades are shown as dashed lines.

[0071] Adjustable force is applied via piston **364** to the shear blades by hydraulic pressure using clean turbine oil, i.e., the process solvent, that is conducted up the drive shaft **46** from a hydraulic swivel just below the lower shaft bearing. Piston **364** moves inside GGB-CBM graphite/bronze bearings **363** and employs a Graphoil seal **365**. Blade clamp **341** is electrically common with the shear blades, with insulation isolating the blade from the piston **364**. The center of piston **364** is filled with furnace cement to seal it and guide and protect the continuity sensing wire **366**. Wire **366** is bonded to blade **342** and in electrical communication to an ohm meter allows electrical conductance measurement to be used to indicate physical contact between the shear blade and **141** wall. Hydraulic pressure against piston **364** is adjusted until the blade **342**, **343** hydroplanes on a surface layer in which a high strain rate is present. At high hydraulic pressure and low or nil rotation speed, carbon build-up that forms a semi-conductive path between the blade **342**, **343** and wall **141** can be detected. If a dedicated scraper is not installed, the shear blades **342**, **343** can be employed to scrape carbon from the wall by reversing the direction of rotation for a few revolutions until conductivity is reduced to indicate the absence of a carbon layer.

[0072] FIGS. **5A** and **5B** illustrate views down a cylindrical reaction vessel **140** of slightly different implementations of the shear plate assembly of FIG. **2**. As illustrated, a six-bolt flange **144** is welded to the central shaft **146** with five sets of arms **148** supporting shear-plates **142** and scrapers **143**. Other than slight variations in the construction of frame **147** and wheels **149**, the assemblies are essentially identical.

[0073] FIG. **6** is a cross-section the bottom portion of an embodiment of a reactor vessel showing how the shear plate drive shaft **146** is coupled to the bottom scraper shaft **451** by a simple spline which includes two spline teeth **453** bolted to the scraper shaft **451**. The teeth **453** move vertically in slots **452** milled in drive shaft **146**, allowing the bottom scraper assembly to be withdrawn upward through the top of the reactor vessel. The overlap between the two shafts **146** and **451** of at least five diameters ensures that the shafts are concentric and that axial bearing **455** can maintain concentricity of the lower end of the shear plate assembly. Axial shaft bearing **455** is retained within bottom cap **457** which is bolted to reactor vessel bottom plate **460**. Bottom plate **460** is attached to the lower end of the reactor vessel via Marman clamp **462**. Solid materials are swept by bottom scrapers **456** to outlet **459**.

[0074] Thrust bearing **445**, which is attached via screws to the bottom of plate **454**, bears the weight of the bottom scraper assembly and maintains the bottom scrapers **456** in light contact with the upper surface of bottom plate **460**. The heat exchanger shell fitting **458** provides the outlet from heat exchanger shell **136** for counter-flowing primary heat transfer fluid. As previously described, steel baffles **137** are brazed to the outer surface of the cylindrical reaction vessel to guide heat transfer fluid to the outer surfaces of the reactor vessel.

[0075] FIG. **7** provides a perspective view of the bottom portion of a reactor vessel shown in FIG. **6**, with the same components identified by the same reference numbers. Rotation in this view is clockwise, as indicated by the arrows on arms **148**. The axis of bottom scraper **456** is offset by plate **454** so that solid material is forced toward the outlet **459**.

[0076] A key aspect of the inventive approach is the combination of the mixing provided by rotating shear plates and the close proximity of that mixing to the heated surface of the reactor

vessel. This combination efficiently conducts endothermic pyrolysis into the process fluid.

[0077] Much higher shear strain rates can be produced with less power via Couette flow than by other mechanical means such as impellers, paddles, or propellers. To illustrate, one can compare how viscous vs. hydrodynamic drag depends on velocity. (It is noted that sound or ultrasound achieves high strain rates but at low transduction efficiency (~10%). Furthermore, magnetic and electrical components do not perform well, or even survive, at elevated temperatures.)

[0078] The hydrodynamic drag force on the shear-plate and scraper assembly rotating at angular velocity ω (rad/sec) = $2\pi f$. sub. $r = 2\pi$ rpm/60 near the cylindrical wall of a reactor vessel of radius r with tangential velocity v . sub. $t = \omega r$ is

F . sub. $hy = C$. sub. DQA . sub. f where the dynamic pressure $Q = \frac{1}{2} \rho v$. sub. t . sup. 2 .

The drag coefficient C . sub. D depends on the Reynolds number $Re = \rho v$. sub. t l / μ which can be estimated to lie in the range $0.5 < C$. sub. $D < 1.5$ when $(102 < Re < 105)$. μ is the absolute viscosity and l is a characteristic length of the body (here, custom-character 0.1 m).

[0079] The viscous drag between a shear plate with area A . sub. s (here ~10 cm × 5 m = 0.5 m²) and reactor wall is

$$[00003] F_{vi} = \left(\frac{v_t}{r} \right) A_s,$$

where $\delta r \sim 2.5$ mm is the shear gap between plate and the reactor wall. Friction drag F . sub. fr on the scraper blade loaded with a light spring force F . sub. $sp \sim 22$ N and further loaded by centrifugal force acting upon the mass (m . sub. $s = 2.32$ kg) is

$$[00004] F_{fr} = F_{n_k} \quad \text{where} \quad F_n = F_{sp} + m_s r^2$$

and the coefficient of kinetic friction $\mu_k \sim 0.2$ for a bronze scraper on a stainless steel tank wall. A simulation was run using the system parameters listed in Table 3. With hydrodynamic drag, Couette flow (viscous) drag and scraper friction drag accounted for, the simulation results, including total torque $\tau = r(F$. sub. $hy + F$. sub. $vi + F$. sub. $fr)$ and power $P = \omega \tau$, are shown in Table 4.

TABLE-US-00003 TABLE 3 Parameter Value # of arms 2 (scraper + shear plates) Radius 0.381 m Length 5.0 m Height 2.0 cm C_D 1.0 Couette gap 2.0 mm Plate width 10.0 cm Scraper mass 2.32 kg Spring 22.0 N Friction coefficient 0.20 Fluid density 1200 kg/m³ (Mobiltherm 600 @50° C.) Fluid viscosity 3.03e-02 N * s/m² (Mobiltherm 600 @50° C.)

[0080] The (viscous) drag force F . sub. vi on shear plate assembly increases linearly with rpm while hydrodynamic force F . sub. hy increases as the square of the rotational speed so that hydrodynamic (pressure) drag F . sub. hy dominates at all rotation speeds and precludes operation at speeds much above about 60 rpm.

[0081] For a CDP plant with m . sub. $feed = 100$ Tons/day using a solvent oil to feed ratio of 2:1 and a process fluid density ρ . sub. $pf \sim 1.0$ Ton/m³ (since ρ . sub. $oil \leq 1$ Ton/m³ and ρ . sub. $feed \geq 1$ Ton/m³, the volume flow rate of process fluid V . sub. $p = 3$ m . sub. $feed / \rho$. sub. $pf = 0.21$ m³/min.

[0082] The volume of process fluid in the reactor V . sub. $r = (\pi/4) D$. sup. $2 L = 2.95$ m³ so that a mean residence time of a parcel of fluid in a reactor would be V . sub. r / V . sub. $p = 2.95 / 0.21 = 14.0$ min.

[0083] For a liquid phase pyrolysis reactor of the scale suggested herein ($D = 0.75$ m) × $L = 5$ m), the volume of fluid per unit of time that is processed between the reactor wall and shear plates (two per reactor) is $\dot{V} = v$. sub. $t A$. sub. gap . For example, if 40 rpm is chosen from Table 4 below with tangential velocity v . sub. $t = 1.6$ m/s and A . sub. $gap = (2 \text{ mm}) \times (5 \text{ m}) \times 2 \text{ plates} = 0.025$ m², then $\dot{V} = 0.04$ m³/s = 2.4 m³/min.

[0084] The ratio ρ . sub. f the volume rate of process fluid that passes between the shear plates and reactor walls and the volume rate at which it flows through the plant $\dot{V}$. sub. $r / \dot{V}$. sub. $p = 11.4$ per reactor. Given a minimum of three reactors per plant, each parcel of process fluid will be subjected to high shear rate near a heated reactor wall an average of 34 times. (Note that this is a low estimate since the process fluid volume decreases as product leaves as vapor to the distillation columns.)

TABLE-US-0004 TABLE 4 Speed v.sub.t Re Q F.sub.hy F.sub.vi F.sub.fr Sdot Power (rpm) (m/s) (#) (Pa) (N) (N) (N) (1/s) (W) 5 0.199 790 23.9 6.57 3.02 8.9 99.7 3.69 10 0.399 1,580 95.5 26.3 6.04 9.19 199 16.6 20 0.798 3,160 382 105 12.1 10.4 399 102 40 1.60 6,320 1,530 420 24.2 15 798 733 50 1.99 7,900 2,390 657 30.2 18.5 997 1,410 60 2.39 9,480 3,440 946 36.3 22.8 1,200 2,400 100 3.99 15,800 9,550 2,630 60.4 47.6 1,990 10,900 200 7.98 31,600 38,200 10,500 121 164 3,990 86,100

[0085] This industrial-scale $\{\dot{V}\}$ is achieved at a strain rate $\{\dot{\epsilon}\}=798/\text{s}$ ('Sdot' in Table 4), which is equal to the maximum and roughly twice as high as the mean value over the in-plane flow field achieved by the best of the CSTR impellers described by S. Vlaev, et al. (*J. Eng. Sci. Tech.*, 2(2):177-87, 2007).

[0086] CDP processing of feedstock using the inventive reactor includes preparation of the process fluid mixture. FIG. 9, discussed below, illustrates an exemplary set-up for such processing.

Referring to FIG. 8, in step **802** fresh catalyst is mixed thoroughly with recycled solvent oil at modest temperature ($\sim 100^\circ \text{C.}$). In step **804**, the oil/catalyst dispersion is added to the incoming charge of feedstock in a low speed mechanical mixer to make a slurry.

[0087] For improved efficiency, a processing set-up that includes two (or more) input tanks (see, e.g., tanks **562** in FIG. 9) allows alternating between the tanks to supply a continuous flow of reactants. While a continuous flow of reactant material is being fed into the first input tank, the pre-processing preparatory steps can be performed in another of the two or more tanks: [0088] Step **806**: a slurry pump is used to transfer the charge into the input tank. [0089] Step **808**: the charge is de-aired via a liquid ring vacuum pump. [0090] Step **810**: the charge is brought up to -200°C. for introduction to the first stage reactor. [0091] Step **812**: chamber ullage is purged by evolved steam and other volatile gases and vapors. Column overhead gas or naphtha can be injected to complete the purge if oxygen sensors signal the necessity.

[0092] After pre-processing has been completed, in step **814** the process fluid mixture is further heated to bring up to process temperatures and is transferred into the reactor (first stage if a multi-stage set-up) for processing in step **816**. The reactor motor is activated to use the combination of stirring, shear forces and heat to process the mixture as further described herein for a pre-determined period of time. As described below with reference to FIG. 9, multiple reactors may be connected in series to allow processing at different temperatures and removal of extracted components at different stages.

[0093] The cylindrical reaction vessels disclosed herein with their concentric external heat exchange shells and internal surfaces swept with rotating shear plate/scrapper blades are versatile components for a wastepaper and plastic pyrolysis plant for many reasons. Advantages include: (1) they can be deployed in series so that their temperatures can be increased in stages; (2) the residence time in each temperature stage can be adjusted by (a) changing the number of reactors deployed in parallel in a temperature stage; and/or (b) changing the length of reactor(s) in a temperature stage to preserve the surface to volume ratio needed for heat exchange; and (3) a plant composed of interchangeable modules can be tailored to product demand and feedstock composition and availability.

[0094] FIG. 9 provides a schematic for one possible implementation of a CDP plant **500** incorporating the inventive shear plate reactors. Three thermal stages, i.e., low, medium and high are shown (four, if the feed/lock stage is counted) with equal length of shear plate reactors **567L**, **567M** and **567H**, respectively, in each stage. The number of temperature stages, reactor length, and number of reactors per temperature stage can vary. For example, if diesel fuel is favored as a product, there may be multiple high temperature reactors in series and parallel combinations, all in the diesel boiling range (up to $\sim 390^\circ \text{C.}$) to increase the residence time of process fluid at this stage.

[0095] The use of combined heat and power (CHP) is important, if not necessary, for the economic viability of waste plastic processing. A turbine-driven electric generator (genset) is preferred over diesel since the latter rejects the majority of waste heat to its liquid cooling system at too low

temperature to be useful. Accordingly, the CDP plant **500** depicted in FIG. **9** employs a molten salt transport circuit **502** (gray lines) and process steam heat transport circuit **504** (dashed lines). The solid black lines designate the solvent+polymer circuit **506**. Turbine exhaust first heats molten salt in exchanger **566** and then in series generates process steam in boiler **572**.

[0096] On start-up from ambient temperature the CHP gas turbine running on stored compressed gas **573**, raises process steam that is directed to preheat exchanger **564s** in the salt tank **563** (to which the salt circuits drain on shut-down) and to exchanger **564p** to preheat the process solvent and thus the reactors as the solvent circulates. Pre-heat of still re-boiler **569** and still column **568** is accomplished by condensing process fluid vapor. In operation, one molten salt circuit heats the still re-boiler **569** and another circuit heats high, medium and low temperature reaction vessels **567H**, **567M** and **567L** in series via concentric, baffled heat exchange shells **577**.

[0097] Ground and sorted plastic is fed into hopper **561** and mixed with catalyst and solvent in feed lock tanks **562**, which may work alternately to provide a continuous flow of plastic slurry for processing. With mild agitation, the charge is de-aired via a liquid ring vacuum pump before steam if admitted to tank ullage where pressure is maintained above ambient as steam admitted to the concentric heat exchange shell heats the charge to 120-140° C. for introduction to the first reactor thermal stage.

[0098] The process fluid (solvent and solute polymers) progresses through the thermal-staged shear plate reactors **567H**, **567M** and **567L** with residence times in each thermal stage adjusted as previously described by changing the vertical lengths of the reactors and the number of reactors held at the same temperature by causing the molten salt to flow in parallel through their heat exchange shells.

[0099] Vapors from each thermal stage are taken via insulated tubes **576** to different vertical heights of the distillation column **568** that matches their respective temperatures. Still “bottoms” are returned via **571** to recycle co-distilled solvent as well as high-boiling polymer fragments for another pass through the high temperature reactor **567H**. Distillation “cuts” are recovered as a portion of the liquid condensate from condensers at **584** with the balance returned to the column as reflux.

[0100] Process fluid is removed from the high temperature stage and processed through a hydro cyclone **579** with underflow containing coarser particles removed at a rate that maintains the fluid levels in the reactors. The cyclone overflow circulates (via route **570**) through the lower temperature reactor shells in order to exchange (recuperate) heat with the incoming fluid before recycling to the feed lock/pre-heaters **562**. A decanting centrifuge or screw press **580** then processes the cyclone underflow after which the wet solids are subjected to a counter flow of steam to co-distill solvent and other adsorbed material from the surface of the coke (and coke adhered to “spent” catalyst). Steam and co-distilled species are passed to condenser **581** with air or water cooling, then to a separator **582** from which water is returned to boiler feed and the hydrocarbon phase recycled to the process fluid.

[0101] The co-distillation apparatus is a horizontal tube having a concentric screw conveyor with slide valves for inlet and exit of coke such that the tube can be sealed for the periodic circulation of steam. The last step of coke clean-up to render it suitable for use as a soil amendment or other application that is exposed to the environment avoids CO₂ production from coke that is burned from the catalyst as standard oil refinery practice in fluidized bed crackers.

[0102] The inventive approach introduces a number of improvements into a CDP system design by placing reactors in series with increasing temperatures: [0103] Moisture and volatile components of the feed material are removed at low temperature, thus saving the energy that would have been used to increase their temperature to that of a single hot reactor. [0104] Fresh catalyst can be added between stages. [0105] Since new cold material is not being added to a single hot reaction vessel, temperature can be much more uniform in each stage. [0106] More efficient use of a heat source by counter-flow of heat transfer fluid with the process fluid as it progresses through the stages. Spent

catalyst and process fluid also counter flow with incoming process fluid. [0107] Efficiency of the subsequent distillation process is improved: Vapor drawn off lower temperature stages can be routed either through separate smaller distillation column(s) or to a single column but at a location higher up the column that matches lower stage's vapor exit temperature. [0108] Compared to a single large reaction vessel, multiple reaction vessels connected in series, while not a true continuous, i.e., non-stop, process, can easily approach the effect of a continuous process by controlling the residence time of reactants within the different reactors.

[0109] Regarding the final point above, a single mixed reaction vessel is inherently a batch process in which the desired reactions are given enough time to progress sufficiently toward completion before addition of a new charge. However, if reactants, catalyst and solvent are continuously added to the contents of a single reaction vessel then, to balance the build-up of spent catalyst, by-product carbon and non-reactive material, some of the reactor's contents must be continuously removed. An adverse consequence of this removal is that a proportionate amount of recently added and as-yet unreacted feedstock will also be removed from the process stream.

[0110] Given that for tanks designed with the same pressure margin of safety, the same mass of tank material will contain the same internal volume whether it is processed within one tank of large diameter or many smaller diameter tanks, it is reasonable to compare several smaller diameter tanks in series, to a single large diameter tank.

[0111] For the sake of illustration, suppose that reaction vessels are being continuously supplied with reactants and catalyst such that 10% of un-reacted feed material and catalyst “short-circuits” the process, finding its way to the exit without reacting.

[0112] With three tanks in series having the same mixing power per unit volume as the single tank and thus the same un-reacted fraction of 10%, the net un-reacted fraction for the output of the three reactors in series will be the product $(0.1)^3 = 0.001$ or 0.1% (i.e., the conditional probability of three successive survival events). This advantage (0.1% un-reacted vs. 10%) for the smaller tanks in series applies where mixing controls the reaction rate. When heating controls the rate, the smaller tanks in series have an even greater advantage over a single tank being used in a (pseudo-) continuous process since the smaller tanks have more area of thinner walls in contact per unit volume of process fluid.

[0113] FIG. 10 illustrates an embodiment with concentrated solar troughs. In this case the exhaust gas is routed through ‘recuperator’ heat exchanger to increase the efficiency of the turbine rather than to heat the chemical process.

[0114] The inventive approach introduces a number of improvements into a CDP system design by placing reactors in series with increasing temperatures: [0115] Moisture and volatile components of the feed material are removed at low temperature, thus saving the energy that would have been used to increase their temperature to that of a single hot reactor. [0116] Fresh catalyst can be added between stages. [0117] Since new cold material is not being added to a single hot reaction vessel, temperature can be much more uniform in each stage. [0118] More efficient use of a heat source by counter-flow of heat transfer fluid with the process fluid as it progresses through the stages. Spent catalyst and process fluid also counter flow with incoming process fluid. [0119] Efficiency of the subsequent distillation process is improved: Vapor drawn off lower temperature stages can be routed either through separate smaller distillation column(s) or to a single column but at a location higher up the column that matches lower stage's vapor exit temperature. [0120] Reaction vessels connected in series can be operated as a continuous process in which the residence time of reactants can be controlled. [0121] (A single mixed reaction vessel is inherently a batch process. Otherwise, if reactants, catalyst and solvent are continuously added to the contents of a single reaction vessel then, to balance the build-up of spent catalyst, by-product carbon and non-reactive material, some of the reactor's contents must be continuously removed. An adverse consequence of this removal is that a proportionate amount of recently added and as-yet unreacted feedstock will also be removed from the process stream.)

[0122] As will be apparent to those of skill in the art, different combinations of the various elements and components described herein may be made to achieve the intended goal of providing an effective and efficient system and method for recycling of plastics and other hydrocarbons. Accordingly, the illustrative examples provided herein and shown in the drawings are not intended to be limiting.

Claims

1. A reactor for liquid phase catalytic pyrolysis of polymers in a slurry of polymeric particles, solvent oil, and a catalyst, the reactor vessel comprising: a cylindrical reactor barrel having an inlet, an outlet, a top plate, a bottom plate and an inner wall configured for transferring heat to the slurry; a central shaft supported concentrically and rotatably between the top plate and the bottom plate; a drive motor configured for driving rotation of the central shaft; at least one arm set extending radially from the central shaft, each arm of the at least one arm set having a distal end and a compressible spring assembly configured to radially bias the distal end toward the inner wall; and a shear plate supported perpendicularly on the distal end of each arm, the shear plate having a plate surface with a plate length and a plate area, the plate surface extending parallel to the inner wall to define a shear gap between the plate surface and the inner wall, wherein the shear gap is dimensioned to induce viscous drag at a predetermined strain rate within the slurry of approximately $10^{2.2}$ to $10^{3.3}$ per second between the plate surface and the inner wall.
2. The reactor of claim 1, where the distal end of each arm further supports a scraper configured to remove solids deposited on the inner wall, the scraper having a scraper length corresponding to the plate length.
3. The reactor of claim 1, wherein the at least one arm set comprises a plurality of arm sets spaced apart along a length of the central shaft, and wherein one arm of each of the plurality of arm sets supports a single shear plate, wherein the plate length is substantially equal to a barrel length.
4. The reactor of claim 1, further comprising a heat exchange shell surrounding the reactor barrel, the heat exchange shell connected to a heat exchange circuit and configured to flow a heat exchange fluid over an outer surface of the reactor barrel.
5. The reactor of claim 1, wherein a trailing edge of the shear plate is curved away from the inner wall to diffuse fluid flow and direct the slurry away from the inner wall.
6. The reactor of claim 1, further comprising: a frame disposed on the distal end of each arm; at least one wheel rotatably supported on the frame, the at least one wheel configured to contact the inner wall and cooperate with the spring assembly to control the shear gap in case of deviations in one or more of reactor barrel roundness and acentricity of the central shaft.
7. The reactor of claim 1, further comprising an electrically-conductive wire extending through each arm, the wire having a distal end electronically connected to the shear plate and a proximal end connected to a conductivity detector configured to indicate physical contact between the shear plate and the inner wall.
8. The reactor of claim 1, wherein an aspect ratio between a barrel length and a barrel diameter of the reactor barrel is within a range of 5:1 and 10:1.
9. The reactor of claim 1, wherein the drive motor is configured to drive rotation of the central shaft at 40 to 60 rpm.
10. The reactor of claim 1, wherein the shear gap is approximately 2 mm to 3 mm.
11. The reactor of claim 1 in a processing facility comprising a plurality of reactors connected in series, wherein each of the plurality of reactors is configured to operate at a different processing temperature, wherein processing temperatures progressively increase from a first reactor in the series to a final reactor in the series.
12. An improved reactor for liquid phase catalytic pyrolysis of polymers in a slurry of polymer particles, solvent oil, and a catalyst, the improvement comprising a shear plate assembly disposed

on a rotatable shaft concentrically disposed within a cylindrical reactor vessel, the shear plate assembly comprising a shear plate disposed perpendicularly on an arm extending radially from the shaft, parallel to and at a spacing from an inner vessel wall surface to define a shear gap between a shear plate surface and an inner wall of the reactor, wherein the shear gap is dimensioned to induce viscous drag within the slurry between the plate surface and the inner wall at a strain rate of approximately $10^{2.2}$ to $10^{3.3}$ per second when the shear plate and arms rotate within the vessel.

13. The improved reactor of claim 12, further comprising a scraper is disposed on a leading edge of the shear plate to remove material deposited on the inner vessel wall surface.

14. The improved reactor of claim 12, further comprising a heat exchange shell surrounding the reactor vessel, the heat exchange shell connected to a heat exchange circuit and configured to flow a heat exchange fluid over an outer surface of the reactor vessel.

15. The improved reactor of claim 14, wherein the heat exchange circuit comprises a molten salt circuit.

16. The improved reactor of claim 12, wherein the arms comprise a plurality of arm sets spaced apart along a length of the rotatable shaft, and wherein one arm of each of the plurality of arm sets supports a single shear plate having a shear plate length corresponding to a length the reactor vessel.

17. The improved reactor of claim 12, wherein a trailing edge of the shear plate is curved away from the inner vessel wall surface to diffuse fluid flow and direct the slurry away from the inner vessel wall surface.

18. The improved reactor of claim 12, wherein each arm comprises a piston configured to apply variable pressure against a distal end of the arm.

19. The improved reactor of claim 12, further comprising an electrically-conductive wire extending through each arm, the wire having a distal end electronically connected to the shear plate and a proximal end connected to a conductivity detector configured to indicate physical contact between the shear plate and the inner wall surface.

20. The improved reactor of claim 12, wherein the shear gap is approximately 2 mm to 3 mm.
